Do Now · retrieval

Software development: practical testing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a prototype?

Q2

Why might a tester run tests at each sprint?

Q3

What does AND require for a compound condition to be true?

Q4

For valid ages 12–16 inclusive, which four values test either side of both boundaries?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

State the valid range — The specification permits whole-number marks from 0 through 100 inclusive.

Core idea 2

Choose targeted tests — Normal value 50; valid boundaries 0 and 100; invalid just beyond boundaries −1 and 101.

Core idea 3

Write expected results — A useful test log records input, expected outcome and actual outcome; a mismatch is evidence of a fault.

Key visual · complete concept

TEST RECORD

Input	0
Expected	true
Actual	true
Outcome	Pass

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

normal test — Typical valid input.

boundary test — Input at or near a valid limit.

erroneous test — Input outside permitted requirements.

expected result — Predicted correct output for a test.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

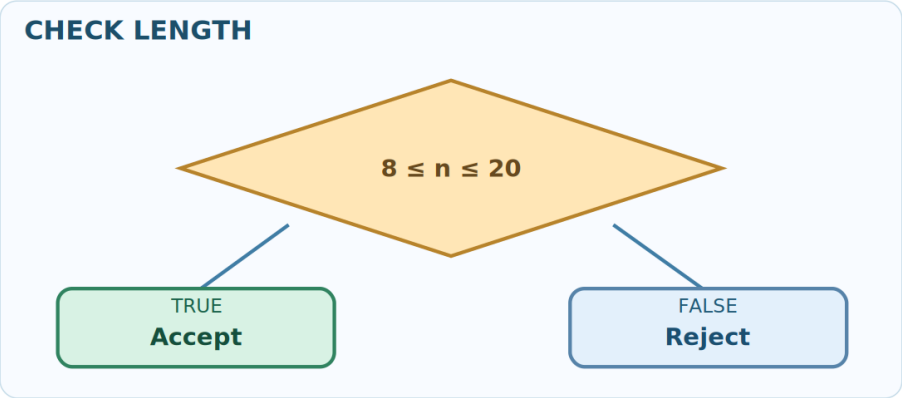
Problem

Design tests for a password length from 8 through 20 characters inclusive.

Follow the worked method

1. Predict first — Test just below and at the lower limit: 7 rejected, 8 accepted.
2. Model and explain — Also test a normal 14, upper boundary 20, and 21 rejected.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Lower boundary 1 is accepted while 0 is invalid.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A whole-number temperature must be 10–40 inclusive. Give a normal, a lower-boundary, an upper-boundary and an erroneous test.

Question 2

Write C# to validate a score from 1 to 9 inclusive and give four expected-result test pairs.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A developer tests values 20,30,40 for a permitted range 20–40. Explain why this is insufficient and propose a stronger suite.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Testing only typical valid inputs proves that validation rejects out-of-range values.

Use key vocabulary from memory

1. Explain normal test in your own words.

2. Explain boundary test in your own words.

3. Explain erroneous test in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.